

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide $K_4Fe(CN)_6$. After filtering, the precipitate is dried and weighed. The reactions which occur are $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$ $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$ If a 2 gram sample of rock yields 0.969 gram of $Zn_2Fe(CN)_6$, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight $Zn_2Fe(CN)_6 = 342.6$.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<tool_call> reasoning process here

We are given :

- A 2 gram sample of rock yields 0.969 grams of $Zn_2Fe(CN)_6$.
- We are to find the percentage of Zn in the original rock sample .
- The reactions are :
1. $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$
2. $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$

From the second reaction , 2 moles of $ZnCl_2$ produce 1 mole of $Zn_2Fe(CN)_6$.

So , 1 mole of $Zn_2Fe(CN)_6$ contains 2 moles of Zn .

We are given :

- Molecular weight of $Zn_2Fe(CN)_6 = 342.6$ g/mol
- Atomic weight of Zn = 65.4 g/mol $\rightarrow$ so 2 moles of Zn = $2 \times 65.4 = 130.8$ g

So , in 342.6 g of $Zn_2Fe(CN)_6$, the mass of Zn is 130.8 g .

Therefore , the fraction of Zn in $Zn_2Fe(CN)_6$ is :

$(130.8 / 342.6)$

Now , from the experiment :

Mass of $Zn_2Fe(CN)_6$ obtained = 0.969 g

So , mass of Zn in that precipitate = $(130.8 / 342.6) \times 0.969$

Let's compute that :

$130.8 / 342.6 \approx 0.3817$

$0.3817 \times 0.969 \approx 0.3705$ g of Zn

This is the mass of Zn in the 2 gram sample of rock .

So , percentage of Zn in the rock = $(0.3705 / 2) \times 100 = 18.525\%$

Rounded to nearest whole number , this is approximately 19 %

So , the closest answer is :

<answer> 19 % Zn </answer>

Note : The answer choice D is "19 % Zn"

Answer choices :

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Thus , the correct answer is D .

<tool_call> <answer> 19 % Zn </answer>